

FORMULAS FOR REFERENCE

SPHERE	Surface area	$= 4\pi r^2$
	Volume	$= \frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	$= 2\pi rh$
	Volume	$= \pi r^2 h$
CONE	Area of curved surface	$= \pi rl$
	Volume	$= \frac{1}{3}\pi r^2 h$
PRISM	Volume	$= \text{base area} \times \text{height}$
PYRAMID	Volume	$= \frac{1}{3} \times \text{base area} \times \text{height}$

There are 36 questions in Section A and 18 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.

Section A

1. If $A = \frac{h}{2}(a+b)$, then $b =$

A. $2A - ah$.

B. $\frac{2}{h}(A - a)$.

C. $\frac{2A - a}{h}$.

D. $a - \frac{2A}{h}$.

E. $\frac{2A}{h} - a$.

2. Factorize $x^2 - x - xy + y$.

A. $(x - y)(x - 1)$

B. $(x - y)(x + 1)$

C. $(x + y)(x - 1)$

D. $(1 - x)(x + y)$

E. $(1 + x)(y - x)$

3. Simplify $\frac{(a^3b^{-1})^{-2}}{(a^{-1}b^2)^4}$.

A. $\frac{1}{ab^3}$

B. $\frac{1}{a^2b^3}$

C. $\frac{1}{a^2b^6}$

D. $\frac{1}{a^2b^9}$

E. $\frac{a^4}{b^6}$

4. Let $f(x) = 3x^2 + ax - 7$. If $f(-1) = 0$, find $f(-2)$.

A. -27

B. -11

C. -3

D. 1

E. 13

5. If $\begin{cases} y = x^2 - 1 \\ y = 2x - 2 \end{cases}$, then $y =$

A. -4.

B. 0.

C. 1.

D. 0 or 8.

E. -4 or 4.

6. Find the values of x which satisfy both $x + 3 > 0$ and $-2x < 1$.

A. $x > -3$

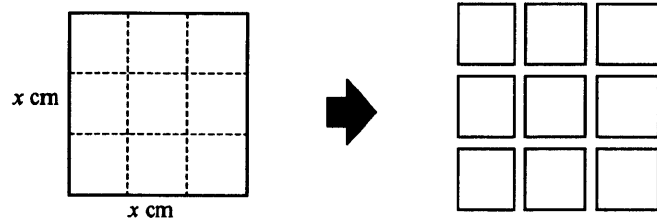
B. $x > -\frac{1}{2}$

C. $x > \frac{1}{2}$

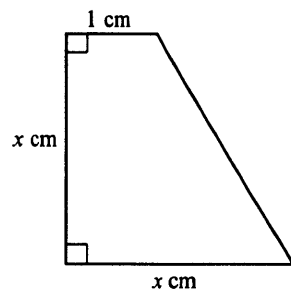
D. $-3 < x < -\frac{1}{2}$

E. $-3 < x < \frac{1}{2}$

7. In the figure, a square of side x cm is cut into 9 equal squares. If the total perimeter of the 9 small squares is 72 cm more than the perimeter of the original square, then $x =$



- A. 6.
B. 8.
C. 9.
D. 12.
E. 18.
8. The figure shows a trapezium of area 6 cm^2 . Find x .



- A. 2
B. 3
C. 4
D. $\sqrt{6}$
E. $\sqrt{11}$

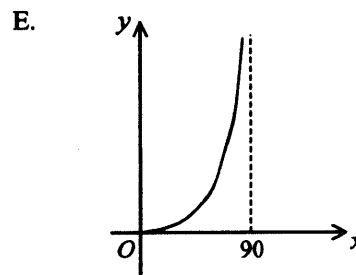
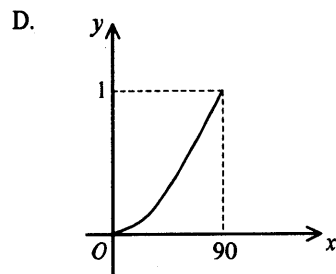
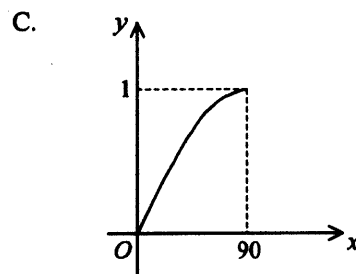
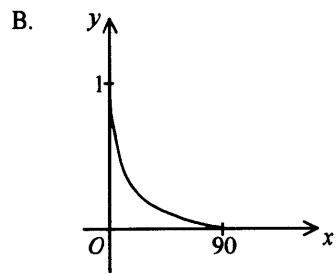
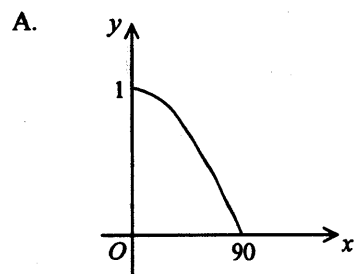
9. Let $f(x) = x^3 - 2x^2 - 5x + 6$. It is known that $f(1) = 0$. $f(x)$ can be factorized as

- A. $(x-1)^2(x+6)$.
B. $(x-1)(x+1)(x+6)$.
C. $(x-1)(x-2)(x+3)$.
D. $(x-1)(x+2)(x-3)$.
E. $(x+1)(x-2)(x-3)$.

10. If $3x^2 + ax + 7 \equiv 3(x-2)^2 + b$, then

- A. $a = -12, b = -5$.
B. $a = -12, b = 7$.
C. $a = -4, b = 3$.
D. $a = 0, b = -5$.
E. $a = 0, b = 19$.

11. Which of the following may represent the graph of $y = \tan x^\circ$ for $0 \leq x \leq 90$?



12. In the figure, $ABCD$ is a rectangle formed by four squares each of area 1 cm^2 . DB is a diagonal. Find the area of the shaded region.

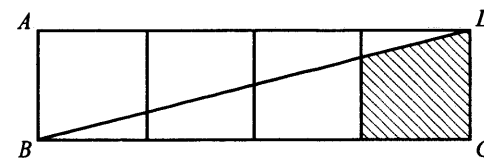
A. $\frac{9}{10} \text{ cm}^2$

B. $\frac{7}{8} \text{ cm}^2$

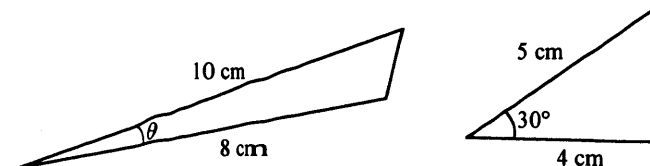
C. $\frac{5}{6} \text{ cm}^2$

D. $\frac{4}{5} \text{ cm}^2$

E. $\frac{3}{4} \text{ cm}^2$



13. In the figure, the areas of the two triangles are equal. Find θ .



A. 7.2° (correct to the nearest 0.1°)

B. 7.5° (correct to the nearest 0.1°)

C. 14.5° (correct to the nearest 0.1°)

D. 15°

E. 30°

14. A man bought two books at \$30 and \$70 respectively. He sold the first one at a profit of 20% and the second one at a loss of 10%. On the whole, he

A. lost 1% .
B. lost 10% .
C. gained 1% .
D. gained 10% .
E. gained 13% .

15. The 1st and 10th terms of an arithmetic sequence are 2 and 29 respectively. The 20th term of the sequence is

A. 56 .
B. 58 .
C. 59 .
D. 60 .
E. 62 .

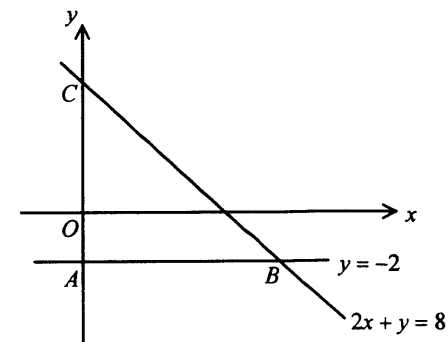
16. Which of the following could be a geometric sequence/geometric sequences?

I. $3, 3^3, 3^5, 3^7, \dots$
II. $9, 99, 999, 9999, \dots$
III. $10, -100, 1000, -10000, \dots$

A. III only
B. I and II only
C. I and III only
D. II and III only
E. I, II and III

17. In the figure, find the area of $\triangle ABC$.

A. 12
B. 15
C. 16
D. 20
E. 25



18. Consider the three straight lines

$$L_1 : 6x + 4y - 3 = 0 ,$$

$$L_2 : y = -\frac{3}{2}x + 4 \text{ and}$$

$$L_3 : 6x - 4y + 3 = 0 .$$

Which of the following is/are true?

I. $L_1 \parallel L_2$

II. $L_2 \parallel L_3$

III. $L_1 \perp L_3$

A. I only

B. II only

C. III only

D. I and III only

E. II and III only

19. In the figure, $ABCD$ is a parallelogram. Find $\angle BDE$.

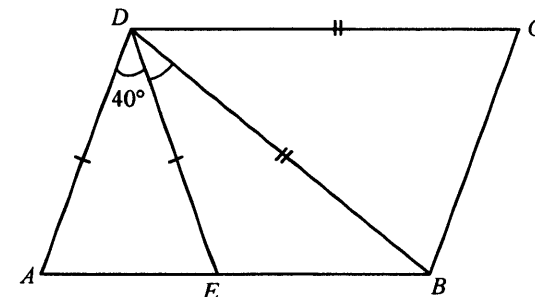
A. 30°

B. 35°

C. 40°

D. 50°

E. 55°



20. In the figure, O is the centre of the circle. $EAOB$ and EDC are straight lines. Find x .

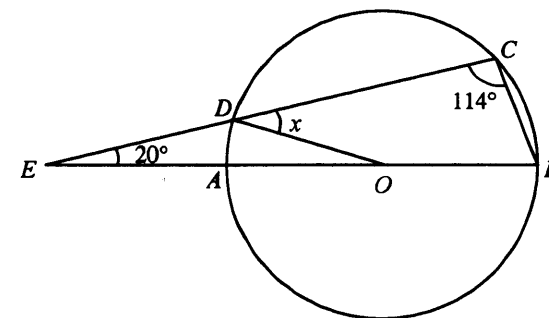
A. 40°

B. 46°

C. 57°

D. 66°

E. 68°



21. Two fair dice are thrown. Find the probability that at least one "6" occurs.

- A. $\frac{1}{3}$
- B. $\frac{1}{6}$
- C. $\frac{5}{18}$
- D. $\frac{7}{36}$
- E. $\frac{11}{36}$

22. A bag contains six balls which are marked with the numbers -3 , -2 , -1 , 1 , 2 and 3 respectively. Two balls are drawn randomly from the bag. Find the probability that the sum of the numbers drawn is zero.

- A. $\frac{1}{30}$
- B. $\frac{1}{10}$
- C. $\frac{1}{5}$
- D. $\frac{1}{3}$
- E. $\frac{1}{2}$

23. $\{x, x+2, x+4, x+6, x+8\}$ and $\{x+1, x+3, x+5, x+7, x+9\}$ are two groups of numbers. Which of the following is/are true?

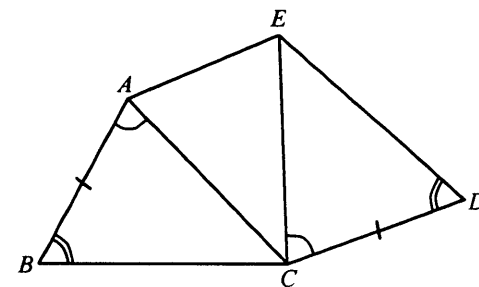
- I. The two groups of numbers have the same range.
- II. The two groups of numbers have the same standard deviation.
- III. The two groups of numbers have the same mean.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

24. In the figure, $AB = CD$, $\angle CAB = \angle ECD$ and $\angle ABC = \angle CDE$. Which of the following must be true?

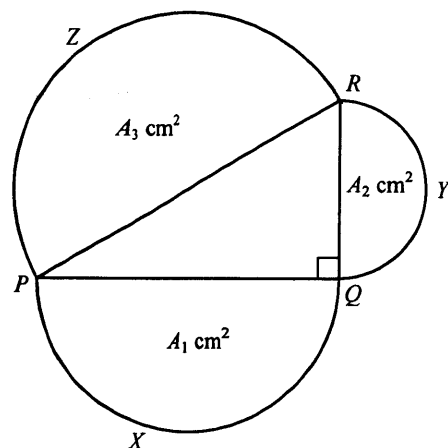
- I. $\triangle ABC \cong \triangle CDE$
- II. $\triangle ABC \sim \triangle EAC$
- III. EAC is an isosceles triangle

- A. I only
- B. III only
- C. I and II only
- D. I and III only
- E. I, II and III



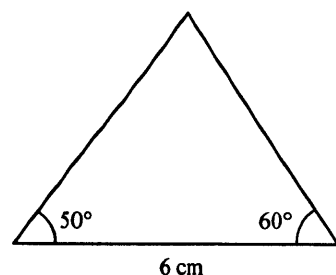
25. In the figure, PXQ , QYR and RZP are semicircles with areas $A_1 \text{ cm}^2$, $A_2 \text{ cm}^2$ and $A_3 \text{ cm}^2$ respectively. If $A_1 = 12$ and $A_2 = 5$, find A_3 .

- A. 13
B. 17
C. 169
D. 13π
E. $\frac{169}{8}\pi$



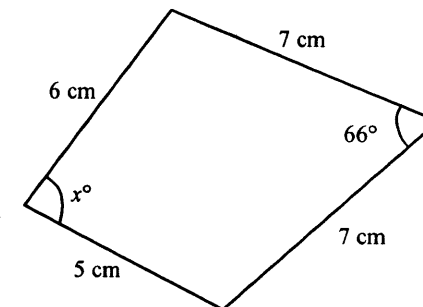
26. In the figure, find the area of the triangle correct to the nearest 0.1 cm^2 .

- A. 7.3 cm^2
B. 10.7 cm^2
C. 12.7 cm^2
D. 15.0 cm^2
E. 19.1 cm^2

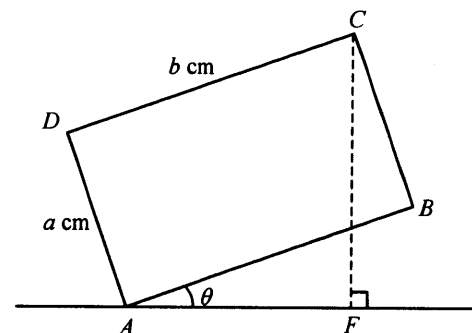


27. In the figure, find x correct to 3 significant figures.

- A. 63.8
B. 78.5
C. 84.5
D. 87.3
E. 89.1

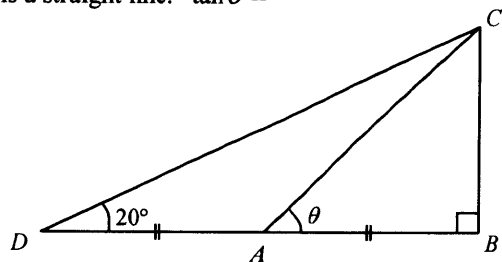


28. In the figure, $ABCD$ is a rectangle. Find CF .



- A. $(a+b)\sin \theta \text{ cm}$
B. $(a+b)\cos \theta \text{ cm}$
C. $(a\sin \theta + b\cos \theta) \text{ cm}$
D. $(a\cos \theta + b\sin \theta) \text{ cm}$
E. $\sqrt{a^2 + b^2} \sin 2\theta \text{ cm}$

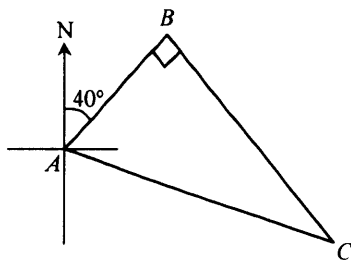
29. In the figure, DAB is a straight line. $\tan \theta =$



- A. $2 \tan 20^\circ$
- B. $\frac{1}{2} \tan 20^\circ$
- C. $\frac{2}{\tan 20^\circ}$
- D. $\frac{1}{2 \tan 20^\circ}$
- E. $\tan 40^\circ$

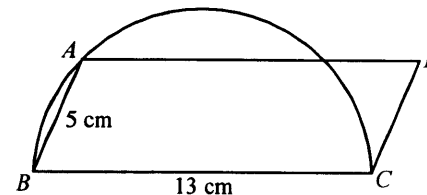
30. According to the figure, the bearing of B from C is

- A. 050°
- B. 130°
- C. 140°
- D. 310°
- E. 320°

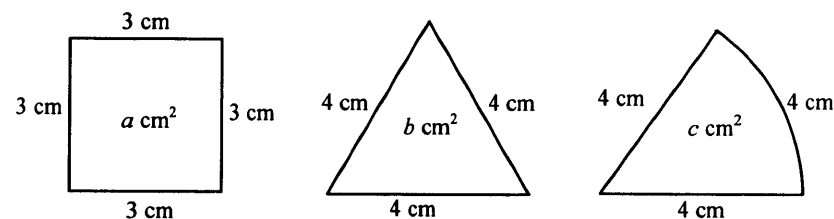


31. In the figure, CAB is a semicircle and $ABCD$ is a parallelogram. Find the area of $ABCD$.

- A. 65 cm^2
- B. 60 cm^2
- C. 52 cm^2
- D. 32.5 cm^2
- E. 30 cm^2



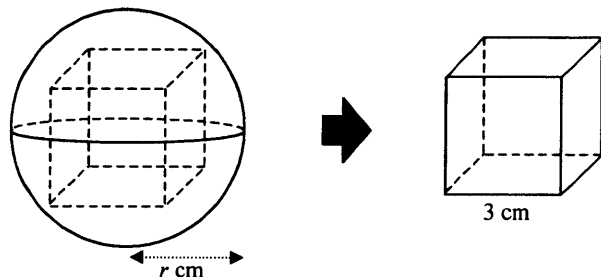
32. The figure shows a square, a triangle and a sector with areas $a \text{ cm}^2$, $b \text{ cm}^2$ and $c \text{ cm}^2$ respectively.



Which of the following is true?

- A. $a > b > c$
- B. $a > c > b$
- C. $b > a > c$
- D. $b > c > a$
- E. $c > a > b$

33. In the figure, a solid wooden sphere of radius r cm is to be cut into a cube of side 3 cm. Find the smallest possible value of r .



- A. $\frac{3\sqrt{3}}{2}$
 B. $\frac{3\sqrt{2}}{2}$
 C. $\frac{3}{2}$
 D. $3\sqrt{3}$
 E. $3\sqrt{2}$

34. If $9a^2 - b^2 = 0$ and $ab < 0$, then $\frac{a-b}{a+b} =$

- A. -2 .
 B. $-\frac{1}{2}$.
 C. 0 .
 D. $\frac{1}{2}$.
 E. 2 .

35. y varies directly as x^2 and inversely as $\sqrt{z}$. If $y=1$ when $x=2$ and $z=9$, find y when $x=1$ and $z=4$.

- A. $\frac{2}{3}$
 B. $\frac{8}{3}$
 C. $\frac{1}{6}$
 D. $\frac{3}{8}$
 E. $\frac{9}{26}$

36. Tea A and tea B are mixed in the ratio $x:y$ by weight. A costs \$80/kg and B costs \$100/kg. If the cost of A is increased by 10% and that of B is decreased by 12%, the cost of the mixture per kg remains unchanged. Find $x:y$.

- A. $1:1$
 B. $2:3$
 C. $3:2$
 D. $5:6$
 E. $6:5$

Section B

37. Simplify $\frac{a}{a+b} + \frac{b}{b-a} + \frac{2ab}{a^2-b^2}$.

A. $\frac{a+b}{a-b}$

B. $-\frac{a-b}{a+b}$

C. $\frac{-a^2+b^2+4ab}{a^2-b^2}$

D. $\frac{a^2+b^2}{a^2-b^2}$

E. 1

38. If $\log(x-a)=3$, then $x=$

A. 10^{3+a}

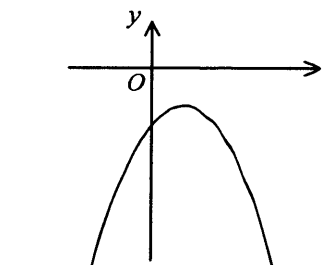
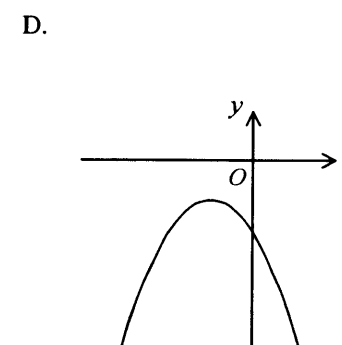
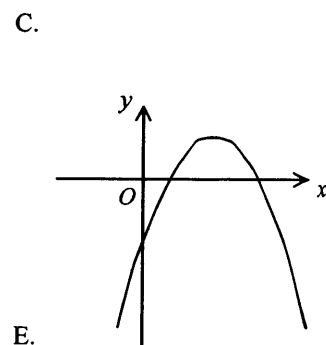
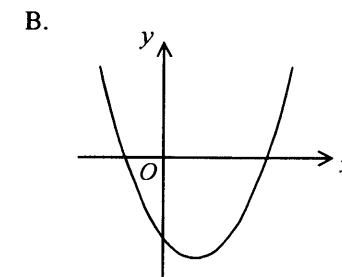
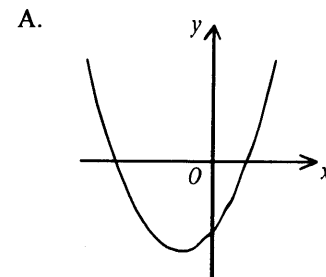
B. a^3

C. $1000a$

D. $1000+a$

E. $30+a$

39. Which of the following may represent the graph of $y = -x^2 + 2x - 3$?



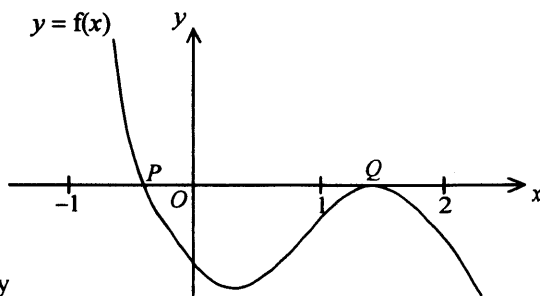
40. If $\left(\frac{\sqrt{5}}{2} + 1\right)x = \sqrt{2}$, then $x =$

- A. $2\sqrt{10} - 2$
- B. $2\sqrt{10} - 4\sqrt{2}$
- C. $2\sqrt{10} + 4\sqrt{2}$
- D. $\frac{\sqrt{10} - 1}{2}$
- E. $\frac{2\sqrt{10} - 4\sqrt{2}}{3}$

41. In the figure, the graph of $y = f(x)$ intersects the x -axis at P and Q only. In order to find a root of $f(x) = 0$ using the method of bisection, which of the following intervals can you start with?

- I. $-1 < x < 0$
- II. $-1 < x < 1$
- III. $1 < x < 2$

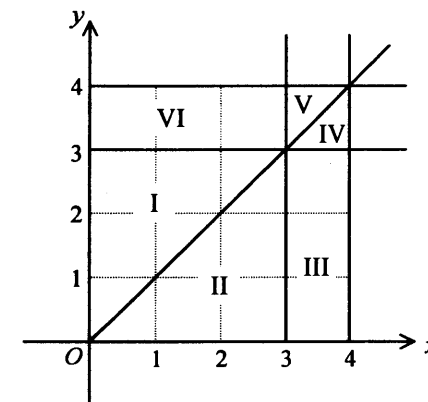
- A. I only
- B. III only
- C. I and II only
- D. I and III only
- E. I, II and III



42. According to the figure, which of the following represents the solution of

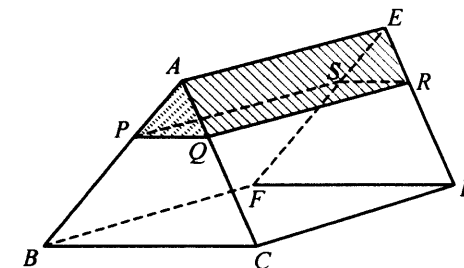
$$\begin{cases} 0 \leq x \leq 4 \\ x \geq y \\ 0 \leq y \leq 3 \end{cases} ?$$

- A. Region I
- B. Region II
- C. Regions I and VI
- D. Regions II and III
- E. Regions II, III and IV



43. In the figure, $ABCDEF$ is a right triangular prism. It is cut into two parts along the plane $PQRS$, which is parallel to the face $BCDF$, and $AP : PB = 2 : 5$. Find $\frac{\text{volume of the prism } APQRES}{\text{volume of the prism } ABCDEF}$.

- A. $\frac{2}{7}$
- B. $\frac{4}{25}$
- C. $\frac{4}{49}$
- D. $\frac{8}{125}$
- E. $\frac{8}{343}$

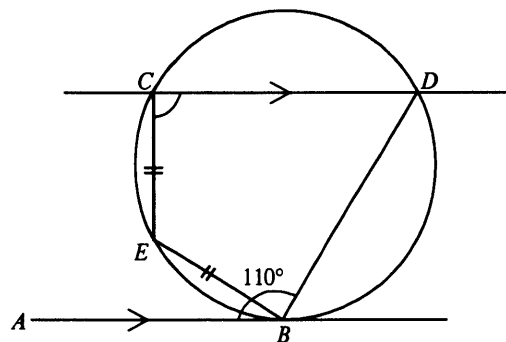


44. π degrees =

- A. $\frac{\pi^2}{180}$ radian .
- B. $\frac{180}{\pi^2}$ radians .
- C. $\frac{\pi}{180}$ radian .
- D. 180 radians .
- E. 1 radian .

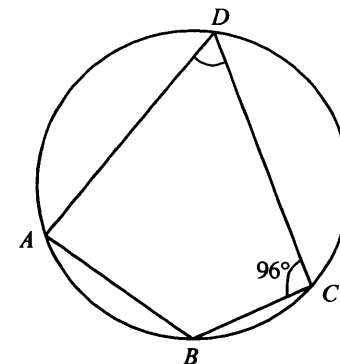
45. In the figure, AB is tangent to the circle at B . Find $\angle DCE$.

- A. 70°
- B. 75°
- C. 90°
- D. 95°
- E. 105°



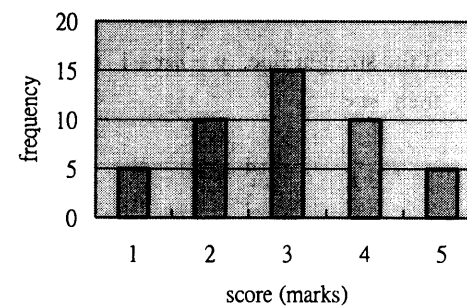
46. In the figure, $\widehat{AB} : \widehat{BC} : \widehat{CD} = 2 : 1 : 3$. Find $\angle ADC$.

- A. 56°
- B. 60°
- C. 63°
- D. 72°
- E. 84°



47. The bar chart below shows the distribution of scores of a test. Find the mean deviation of the scores of the test.

- A. 0 mark
- B. $\frac{8}{9}$ mark
- C. $\frac{2\sqrt{2}}{3}$ mark
- D. $\frac{2\sqrt{3}}{3}$ marks
- E. $\frac{6}{5}$ marks



48. If the centre of the circle $x^2 + y^2 + kx + (k+1)y - 3 = 0$ lies on $x + y + 1 = 0$, find k .

A. $\frac{3}{2}$
 B. $\frac{1}{2}$
 C. 0
 D. -1
 E. $-\frac{3}{2}$

49. If the straight line $y = mx + 1$ is tangent to the circle $(x-2)^2 + y^2 = 1$, then $m =$

A. $-\frac{4}{3}$
 B. 0
 C. $\frac{4}{3}$
 D. 0 or $-\frac{4}{3}$
 E. 0 or $\frac{4}{3}$

50. $A(-1, -4)$ and $B(3, 4)$ are two points. The line $x - y = 0$ cuts AB at P so that $AP : PB = r : 1$. Find r .

A. 3
 B. 2
 C. 1
 D. $\frac{1}{2}$
 E. $\frac{1}{3}$

51. If $\cos \theta = \frac{1}{k}$ and $0^\circ < \theta < 90^\circ$, then $\tan(\theta - 270^\circ) =$

A. $-\frac{k}{\sqrt{1-k^2}}$
 B. $-\frac{1}{\sqrt{k^2-1}}$
 C. $\frac{1}{\sqrt{k^2-1}}$
 D. $-\sqrt{k^2-1}$
 E. $\sqrt{k^2-1}$

48. If the centre of the circle $x^2 + y^2 + kx + (k+1)y - 3 = 0$ lies on $x + y + 1 = 0$, find k .

A. $\frac{3}{2}$
 B. $\frac{1}{2}$
 C. 0
 D. -1
 E. $-\frac{3}{2}$

49. If the straight line $y = mx + 1$ is tangent to the circle $(x - 2)^2 + y^2 = 1$, then $m =$

A. $-\frac{4}{3}$
 B. 0
 C. $\frac{4}{3}$
 D. 0 or $-\frac{4}{3}$
 E. 0 or $\frac{4}{3}$

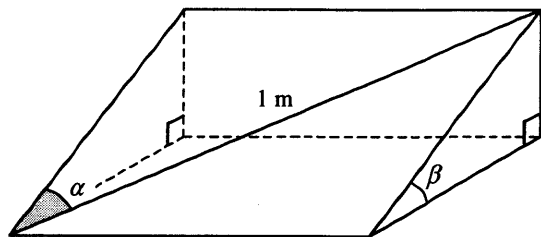
50. $A(-1, -4)$ and $B(3, 4)$ are two points. The line $x - y = 0$ cuts AB at P so that $AP : PB = r : 1$. Find r .

A. 3
 B. 2
 C. 1
 D. $\frac{1}{2}$
 E. $\frac{1}{3}$

51. If $\cos \theta = \frac{1}{k}$ and $0^\circ < \theta < 90^\circ$, then $\tan(\theta - 270^\circ) =$

A. $-\frac{k}{\sqrt{1-k^2}}$
 B. $-\frac{1}{\sqrt{k^2-1}}$
 C. $\frac{1}{\sqrt{k^2-1}}$
 D. $-\sqrt{k^2-1}$
 E. $\sqrt{k^2-1}$

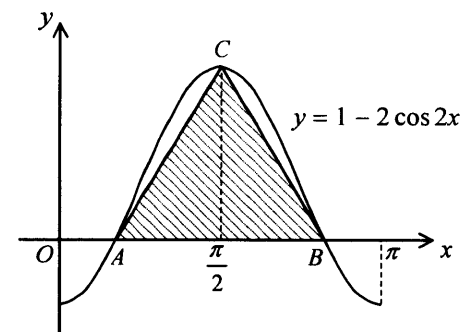
52. The figure shows a right triangular prism. Find its volume.



- A. $\frac{1}{3} \sin^2 \alpha \cos \alpha \sin \beta \cos \beta \text{ m}^3$
 B. $\frac{1}{3} \sin \alpha \cos^2 \alpha \sin \beta \cos \beta \text{ m}^3$
 C. $\frac{1}{2} \sin \alpha \cos \alpha \sin \beta \cos \beta \text{ m}^3$
 D. $\frac{1}{2} \sin^2 \alpha \cos \alpha \sin \beta \cos \beta \text{ m}^3$
 E. $\frac{1}{2} \sin \alpha \cos^2 \alpha \sin \beta \cos \beta \text{ m}^3$

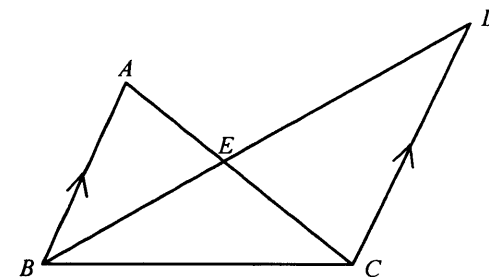
53. In the figure, the area of $\triangle ABC$ is

- A. $\frac{\pi}{3}$
 B. $\frac{2\pi}{3}$
 C. π
 D. $\frac{4\pi}{3}$
 E. 2π



54. In the figure, AEC and BED are straight lines. If the area of $\triangle ABE = 4 \text{ cm}^2$ and the area of $\triangle BCE = 5 \text{ cm}^2$, find the area of $\triangle CDE$.

- A. 4.5 cm^2
 B. 5 cm^2
 C. 6 cm^2
 D. 6.25 cm^2
 E. 9 cm^2



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